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MANAGEMENT REVIEW

Management without theatre.



KPIS

LITTLE'S LAW: WORK IN PROCESS, TIME AND THROUGHPUT

A student's question in Cleveland, a four-step line with and without variability, three numbers that move together, what happens as queues grow, how to count a queue, and a card that turns open work into time.

14 h instead of 8: the time a job spends in a four-step line that holds four jobs, once the process times vary.

HOPP & SPEARMAN, PENNY FAB

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Most reports show how much a team finished. Few show how much is still waiting, and fewer still connect the two with how long a customer waits. Little's law does: three averages, one equation, and no assumptions about how the work is done.

In 1961 John Little proved a formula that teachers of queuing had met again and again: the average number of items in a system equals the rate at which they arrive times the average time each one stays. In Hopp and Spearman's four-step Penny Fab the law holds with and without variability, but the same work in process buys less output. In five Kanban teams, less work in process went with shorter lead times, and also with lower productivity. The card at the end turns a count of open work into a lead time.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Prove it in general

Around 1960 John Little taught queuing at the Case Institute of Technology in Cleveland. He told his class that one formula kept turning up in model after model and seemed very general. After class a student, Sid Hess, asked how hard it would be to prove it in general. Little's answer:

“I guess it shouldn't be too hard.”

01
L
the average number of items in the system, waiting or being served

02
λ
the average rate at which items arrive, per hour or per day

03
W
the average time an item spends in the system

“Famous last words,” he wrote fifty years later. The proof, worked out over summers on Nantucket, appeared in Operations Research in 1961: L = λW, whatever the pattern of arrivals, the service times or the order of service. Little's own explanation is simple: a person standing in a queue can be counted, and at the same time is collecting minutes of waiting.

OUR READING

The law does not say how to shorten a queue. It says that the queue, the pace and the wait cannot be managed one at a time.

Sources: John D. C. Little, Operations Research 59(3), 2011; John D. C. Little, Operations Research 9(3), 1961

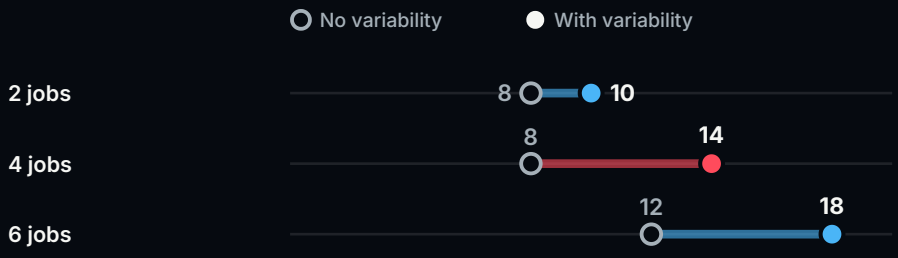
The story and the quotes come from Little's own account for the law's 50th anniversary (2011). The 1961 proof assumes stationary processes; in 2011 Little also proves the law for any finite period.

THE NUMBERS

Same law, longer wait

Wallace Hopp and Mark Spearman teach the law with the Penny Fab: four identical steps in a row, two hours each per penny. A job needs eight hours of work; the line can finish one every two hours. New jobs enter only to keep a set number inside.

HOURS A JOB SPENDS IN THE PENNY FAB, BY JOBS IN THE LINE



Without variability, four jobs fill the line: one leaves every two hours, each after eight; more jobs only add waiting. With variable process times, four jobs bring out 0.286 an hour instead of 0.5, each after 14 hours. In every row, output per hour times hours inside gives back the number of jobs.

OUR READING

Variability does not break the law. It raises the price: the same output needs more work in process, and every extra job is paid for in waiting.

Sources: Joan Tafoya & Tim Skowronski, Intel, IIE Conference, 2008; Wallace J. Hopp & Mark L. Spearman, Irwin/McGraw-Hill, 2000 (via Little (2011), section 4.2; Tafoya & Skowronski (2008))

A teaching example, not measured data: the Penny Fab tables shown by two Intel engineers in 2008, after Hopp and Spearman. Eight hours × 0.5 jobs an hour = the four jobs that fill the line.

THE MODEL

Three numbers, one equation

In operations, Hopp and Spearman put output first: throughput equals work in process divided by cycle time, the time a job spends between release and the end of its route. Little explains why: for an operating manager, output is usually the reason the operation exists, and it is often set from outside, by orders or a forecast.

01

Know two, find the third

The most common use: one number is hard to measure, the other two are not.

02

They move together

More output, shorter waits and less stock cannot be chosen one at a time.

03

Output fixed

Then the only way to cut work in process is to cut the time each item waits.

Each of the three numbers is a measure of performance on its own, Little writes, and managers should consider collecting and displaying all three for whatever stream of items they manage. If something is amiss, it will most likely show in one of them.

OUR READING

A board that shows only what was finished hides two of the three numbers. Count what is open, and the waiting time follows from the arithmetic.

Sources: Wallace J. Hopp & Mark L. Spearman, Irwin/McGraw-Hill, 2000 (via Little (2011), section 4.2; Tafoya & Skowronski (2008)); John D. C. Little, Operations Research 59(3), 2011

Hopp and Spearman's terms as quoted by Little (2011); the three uses and the call to managers are Little's. The reading is the editors'.

WHAT THE RESEARCH SAYS

Less waiting, at a price

The law is exact, but it does not say how much work in process is right. Three reports show what practice adds to it.

01	**Kanban teams, 2018**	Dag Sjøberg studied more than 8,000 work items of five teams in one software company over four years. Less work in process went with shorter lead times, as the literature claims; more work in process also went with higher productivity. No single best limit emerged.
02	**Emergency departments**	Arrivals divided by what one doctor treats give the minimum staff: 10 patients an hour at 2.5 per doctor means 4 doctors. Because queues grow slowly at first and then fast as arrivals near capacity, the rule of thumb reported by Little adds 10–20%.
03	**A server under load, 2010**	In a load test, the queue of requests grew roughly in step with the load, then climbed steeply; above about 18 requests a second, more were essentially dropped. Little reads it as the large queue itself slowing the service.

OUR READING

Full load looks efficient on a capacity chart. In the queue, it is where waiting stops growing in a straight line.

Sources: Dag I. K. Sjøberg, Proceedings of the 12th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM), 2018 (via Abstract at Semantic Scholar); John D. C. Little, Operations Research 59(3), 2011

Sjøberg's study covers one company and shows correlations; productivity was hard to measure. The two other cases are reported by Little (2011) from conversations with the people involved, not as independent studies.

More in the essay High-volume days: the standard under pressure

HOW IT IS MEASURED

Count the queue, find the time

Practitioners always measure over a finite period, Little points out, and over any such period the law holds exactly. A team can measure two numbers and compute the third, or measure all three and check its data.

1. **Draw the boundary**
   Where an item enters, where it leaves, and what counts as one item.
2. **Count what is inside**
   At fixed times, every morning for example; average over the period.
3. **Count what leaves**
   Per day or per hour, over the same period and in the same units.
4. **Divide and compare**
   Inside ÷ leaving = average time inside. Set it against the times you have recorded.

HYPOTHETICAL EXAMPLE, A RETURNS DESK OVER FOUR WEEKS

Open each morning	60 cases on average
Closed per day	20 cases on average
Time inside	60 ÷ 20 = 3 working days

If the system's own timestamps say 1.5 days, a count is wrong, perhaps the cases on hold. The numbers are invented.

Source: John D. C. Little, Operations Research 59(3), 2011

The finite-period proof comes from Little (2011), as does Bill Lovejoy's remark, from a hospital study, that the law is a reality check on data that do not add up. The steps and the example are the editors'.

TOOL OF THE ISSUE

From open work to lead time

One queue, one period. Count what is inside and what leaves, divide, and decide how much open work the team allows before it starts anything new.

01	**Queue and boundary**	where an item enters, where it leaves, which unit
02	**Period**	from and to; the same dates for every count
03	**Work in process**	average count of open items; when and how counted
04	**Throughput**	items finished per day over the period
05	**Lead time = work in process ÷ throughput**	in days; against recorded times and what customers are promised
06	**Limit and check**	most open items allowed; who holds new starts; when we count again

Sources: John D. C. Little, Operations Research 59(3), 2011; Mark L. Spearman, David L. Woodruff & Wallace J. Hopp, International Journal of Production Research 28(5), 1990 (via Abstract reprinted by the Project Production Institute, 2018); Joan Tafoya & Tim Skowronski, Intel, IIE Conference, 2008

A practice proposed by the editors, after Little (2011), the Penny Fab and CONWIP, Spearman, Woodruff and Hopp's pull system that keeps the work in process of a whole line constant.

Open the tool: Delay Analyzer stivencatalyst.com/tools/delay-analyzer/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

A Proof for the Queuing Formula: L = λW
John D. C. Little, Operations Research 9(3), 1961
https://doi.org/10.1287/opre.9.3.383

Little's Law as Viewed on Its 50th Anniversary
John D. C. Little, Operations Research 59(3), 2011
https://doi.org/10.1287/opre.1110.0940

Factory Physics: Foundations of Manufacturing Management, 2nd edition
Wallace J. Hopp & Mark L. Spearman, Irwin/McGraw-Hill, 2000 · via Little (2011), section 4.2; Tafoya & Skowronski (2008)

Factory Physics: A Fast Cycle Time Story (conference slides)
Joan Tafoya & Tim Skowronski, Intel, IIE Conference, 2008
https://beta.factoryphysics.com/wp-content/uploads/2023/02/factory_physics_application_at_intel.pdf

An empirical study of WIP in kanban teams
Dag I. K. Sjøberg, Proceedings of the 12th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM), 2018 · via Abstract at Semantic Scholar
https://doi.org/10.1145/3239235.3239238

CONWIP: a pull alternative to kanban
Mark L. Spearman, David L. Woodruff & Wallace J. Hopp, International Journal of Production Research 28(5), 1990 · via Abstract reprinted by the Project Production Institute, 2018
https://doi.org/10.1080/00207549008942761

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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